

Property Value Effects of Voluntary and Mandatory Flood Buyouts

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Abstract

Flood buyout programs are widely used to reduce long-term exposure to flood risk, yet their effects on surrounding property values remain uncertain. This paper estimates the capitalization of flood buyouts into residential property values using a hedonic pricing model and transaction-level data from Harris County, Texas, covering 2000–2024. We distinguish between voluntary and mandatory buyouts and examine how their effects vary with spatial proximity and the intensity of nearby buyout activity. Using propensity score matching to improve comparability between treated and control properties, we estimate hedonic models across multiple distance bands and buyout counts. We find limited evidence of negative effects associated with voluntary buyouts, with the estimated decline in property values remaining small even at higher levels of exposure. In contrast, greater exposure to mandatory buyouts is associated with larger increases in property values. At the mean level of exposure, voluntary buyouts are associated with an approximately 0.4% decline in property values, compared with increases of 5–7% for mandatory buyouts, depending on distance. To our knowledge, this study provides the first empirical evidence on the property-value effects of mandatory flood buyouts while extending the literature on voluntary buyouts. Overall, the results suggest that the capitalization of buyout activity depends on buyout type and the scale and spatial concentration of acquisitions.

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